

Millicent Ochieng

Research Scientist | Multilingual & Multicultural AI | Equitable AI

✉ millicentochieng950@gmail.com 📍 Nairobi, Kenya 🔗 Website 🌐 LinkedIn 🐙 Github

Profile

I am a Research Scientist at Microsoft Research Africa 📍 and a PhD researcher in Computer Science at the University of the Witwatersrand 📍. My research focuses on multilingual and multicultural AI, where I currently spearhead work on multilingual tokenization to improve linguistic representation and downstream model efficiency across global languages. In parallel, I lead work on cultural alignment, examining how models capture cultural variation across low-resource settings. I have published in top-tier venues including EMNLP, NAACL, and LREC, and I actively contribute to and help shape the African NLP research community through Masakhane 📍 and AfricaNLP 📍. I also serve as WiDS Kenya 📍 Lead Ambassador, leading a community of 400+ members advocating for gender diversity in AI.

My long-term research goal is to develop multilingual AI systems that preserve linguistic and cultural meaning while improving representation, efficiency, and alignment across diverse communities, toward more equitable AI.

Research Interests

- Tokenization & multilingual LLM efficiency
- Multicultural AI alignment across diverse communities
- Globally equitable AI

Publications

08/2026	Toward Cultural Alignment: Human-Centered Evaluation of Multimodal AI Stories Across Five African Communities , <i>EMNLP</i>
03/2026	Reasoning Beyond Labels: Measuring LLM Sentiment in Low-Resource, Culturally Nuanced Contexts 📍, <i>AfricaNLP at EACL</i>
01/2025	IrokoBench: A New Benchmark for African Languages in the Age of Large Language Models 📍, <i>NAACL (Outstanding Paper Award at NAACL 2025)</i>
06/2024	Beyond Metrics: Evaluating LLMs' Effectiveness in Culturally Nuanced, Low-Resource Real-World Scenarios 📍, <i>AfricaNLP at ACL</i>
03/2024	Benchmarking Large Language Models Across Languages, Modalities, Models and Tasks 📍, <i>NAACL</i>
03/2024	AfriMTE and AfriCOMET: Enhancing COMET to Embrace Under-resourced African Languages 📍, <i>NAACL</i>
11/2023	Can Large Language Models Support Medical Facilitation Work? A Speculative Analysis 📍, <i>AfriCHI</i>
10/2023	MEGA: Multilingual Evaluation of Generative AI 📍, <i>EMNLP</i>
10/2022	Participatory Translations of Oshiwambo: Towards Sustainable Culture Preservation with Language Technology 📍, <i>AfricaNLP Workshop at ICLR</i>
06/2022	Language Patterns and Behaviour of the Peer Supporters in Multilingual Healthcare Conversational Forums 📍, <i>LREC</i>

Education

07/2026

Johannesburg,
South Africa**PhD in Computer Science**, *University of the Witwatersrand* [↗](#)

- **Research focus:** Cultural alignment in multilingual language models for African languages, spanning region-conditioned data, modeling, alignment, and pluralistic evaluation.
- **Supervisor:** Prof. Bruce Bassett.

09/2019 – 07/2021

Kigali, Rwanda

MSc. Data Science, *Africa Center of Excellence in Data Science, University of Rwanda* [↗](#)

- **Thesis:** Transfer learning for bean disease classification in precision agriculture.
- **Key Courses:** Machine Learning and Computational Statistics, Big Data Analytics, Programming for Data Scientists, Modeling and Simulation.

01/2014 – 12/2018

Eldoret, Kenya

BSc. Applied Statistics with Computing, *Moi University* [↗](#)

- **Key Courses:** Probability and Statistics, Time Series, Calculus, Linear Algebra.

Professional Experience

08/2021 – present

Nairobi, Kenya

Research Scientist, *Microsoft* [↗](#)

- Multilingual, Multicultural & Equitable AI Research.
 - Spearhead multilingual tokenization research to improve linguistic representation and downstream model efficiency across global languages.
 - Lead work on multicultural AI alignment, examining how models represent cultural variation across diverse communities using human centered methods.
 - Publish in top-tier NLP and AI venues and collaborate across Microsoft and academic research teams.

01/2021 – 07/2021

Cambridge, MA,
United States**Research Intern**, *Institute for Applied Computational Science, Harvard University* [↗](#)

- Applied transfer learning, computer vision, and statistical methods to improve bean disease classification for sustainable agriculture.

07/2019 – 08/2019

Siaya, Kenya

Content Supervisor, *Kenya National Bureau of Statistics* [↗](#)

- Led a team of 10 to ensure accurate data collection for the 2019 Kenya Census.

09/2016 – 12/2016

Mombasa, Kenya

Statistician Intern, *Kenya Ports Authority* [↗](#)

- Applied statistical analysis and forecasting to support port strategic planning.

Skills

Technical Skills

- Python, PyTorch, Hugging Face Transformers, LLMs, Tokenization, Prompt Engineering, Git

Research Methods

- Multilingual & Multicultural NLP, Human-Centered AI, AI Evaluation

Awards

04/2026

Schlumberger Foundation Faculty for the Future Fellowship [↗](#)

- Competitive fellowship supporting women from low- and middle-income countries pursuing PhD research in STEM.

- 01/2026 **WiDS High Impact Ambassador** [🔗](#), *WiDS Worldwide*
 - Recognized for sustained leadership and impact in advancing women in data science and AI.
- 01/2025 **Outstanding Paper Award — IrokoBench** [🔗](#), *NAACL 2025*
 - Recognized for outstanding research contribution in multilingual and African language NLP.
- 09/2019 **IUCEA Masters Scholarship Award** [🔗](#), *University Council for East Africa*
 - Awarded a prestigious STEM scholarship as the top candidate in Kenya.

Research Service

- 10/2025 – Present **AfricaNLP**, *Workshop Organizer & Communications Chair*
 - Co-organized and hosted AfricaNLP at EACL 2026, leading communications and community engagement.
- 05/2023 – Present **ACL**, *Reviewer*
 - Review research submissions in multilingual NLP, LLMs, and related areas.
- 01/2022 – present **Women in Data Science Kenya (WiDS)** [🔗](#), *WiDS Kenya Lead Ambassador*
Nairobi, Kenya
 - Host WiDS events, and serve on WiDS worldwide Ambassador Advisory Council.
- 08/2021 – present **Masakhane** [🔗](#), *Member*
Nairobi, Kenya
 - Advance multilingual NLP research for low-resource African languages.